AMITRIPTYLINE

Fatal in overdose

Anticholinergic

CHEMICAL NAME 3-(10,11-dihydro-5H-dibenzo[a,d]cyclohepten-5-ylidene)-N,N-dimethylpropan-1-amine (hydrochloride)

CLASS Tricyclic antidepressant (TCA)

TRADE NAMES Elavil · Tryptizol · Triptyl · Saroten

THERAPEUTIC USES / INDICATIONS

- Neuropathic pain (low dose) — first-line.
- Migraine and tension-type headache prophylaxis.
- Depression (rarely now; toxicity).
- IBS pain, fibromyalgia, insomnia (low dose, off-label).

ADMINISTRATION

- **PO** at night (sedating).
- Start low and titrate slowly.
- Prescribe limited quantities in at-risk patients (overdose).

DOSAGES

- **Neuropathic pain/migraine:** 10–25 mg at night, increase to 50–75 mg.
- **Depression:** 75 mg/day, up to 150–200 mg (rarely used).
- **Elderly:** 10 mg at night.

PHARMACOKINETICS

- **Onset:** pain 1–2 weeks; depression 2–4 weeks.
- Metabolised (CYP2D6, 2C19) to active nortriptyline.
- **t½:** 10–28 h (nortriptyline 18–44 h).

NURSING MANAGEMENT

- Assess suicide risk; limit supplies.
- Postural BP; falls risk.
- Monitor bowel and bladder function.
- ECG in cardiac patients.

PHARMACODYNAMICS

- Inhibits **noradrenaline and 5-HT reuptake**.
- Blocks **muscarinic, H1, α1** receptors and **sodium channels** (cardiotoxicity).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Dry mouth, constipation, urinary retention, blurred vision.
- Sedation, weight gain.
- Postural hypotension, dizziness.

ADVERSE EFFECTS (SERIOUS)

- Overdose: arrhythmias (QRS widening), seizures, coma, death; treat with sodium bicarbonate.
- Cardiac conduction defects; QT prolongation.
- Confusion/delirium in elderly; angle-closure glaucoma.
- Suicidality; serotonin syndrome.

CONTRAINDICATIONS & CAUTIONS

- Recent MI, arrhythmias, heart block.
- Mania; severe liver disease.
- MAOIs; porphyria.
- **Caution:** glaucoma, BPH, epilepsy, elderly.

KEY INTERACTIONS

- **MAOIs:** hypertensive crisis/serotonin syndrome.
- **Anticholinergics, antihistamines:** additive effects.
- **QT drugs:** arrhythmias.
- **CYP2D6 inhibitors (fluoxetine):** ↑ levels; **alcohol:** sedation.

PATIENT EDUCATION

- Take at night; drowsiness and dry mouth are common.
- Avoid alcohol and driving if drowsy.
- Rise slowly.
- Low doses for pain take 1–2 weeks to work.

KEY POINTS

- TCA — neuropathic pain & migraine at low dose.
- Anticholinergic, sedating.
- 10–25 mg at night.
- Fatal in overdose (cardiotoxic).

SOURCES CHECKED **[786]** Amitriptyline 10 mg Film-Coated Tablets (UK SmPC) **[787]** Amitriptyline Hydrochloride, oral (US label) **[788]** Amitriptyline Hydrochloride, oral (US label, AiPing Pharmaceutical)

Full references p. 432–457